

REDROB HACKATHON · INTELLIGENT CANDIDATE RANKING

Ranking Candidates Like a Great Recruiter

Multi-signal AI ranking: skills × career depth × behavioral availability × honeypot detection

100K

Candidates ranked

53s

Runtime (CPU only)

6

Scoring dimensions

0

External API calls

The Problem with Keyword Matching

What keyword filters see

- ✗ "RAG, Pinecone, LangChain" in skills → ranked #1
- ✗ No check on career evidence or actual usage
- ✗ Ghost candidates: inactive for 8 months
- ✗ Keyword stuffers: expert in 10 skills, 0 months used
- ✗ Consultants-only career treated same as product eng.
- ✗ ~80 honeypot profiles with impossible timelines

What our ranker understands

- ✓ Skill weight = proficiency × endorsements × duration
- ✓ Career: product company vs consulting, role relevance
- ✓ Behavioral multiplier: last active, response rate
- ✓ Honeypot detection: date math, impossible combos
- ✓ Location + notice period fit scoring
- ✓ 100K candidates in 53 seconds, CPU-only

System Architecture

5 scoring stages → 1 ranked output



6 Scoring Dimensions



Skills

35%

Proficiency $\times$ endorsements $\times$ duration_months weighting.
Assessment score bonus (+20%). Off-target skill penalty.
Keyword-stuffing: expert + 0 months = very low weight.



Career History

35%

Product company vs consulting detection.
Role relevance from title + description keywords.
Recency weighting. Pure consulting career: 0.4 $\times$ penalty.



Experience Years

15%

JD sweet spot: 5–9 years. Ideal: 6–8.
Graceful degradation for adjacent bands.
Over-15y: down-weighted (overqualified signal).



Location

10%

Tier 1: Pune/Noida/Delhi NCR. Tier 2: Hyd/Mumbai/Blr.
Notice period scoring: $\leq 30d$ = max, $> 90d$ = 50%.
Non-India without relocation intent: 0.05 base.



Education

5%

Institution tier (1–4) + degree level weighting.
CS/related field bonus (+15%).
Low weight intentionally — JD focuses on ship, not creds.



Behavioral $\times$

Multiplier
0.3–1.2 $\times$

open_to_work, last_active (recency), response_rate,
interview_completion, github_activity,
profile_completeness.
Multiplied onto base score — controls reach-ability.

Honeypot & Fraud Detection

~80 impossible profiles hidden in the 100K pool. We catch them with logic, not gut feel.

⚠ Expert + 0 Duration

PENALTY: 0.6×

≥5 skills claimed as 'expert' with 0 months used → keyword stuffer, not a real expert

⚠ Duration Mismatch

PENALTY: flag

Stated duration_months vs. actual date math disagrees by >18 months per job

⚠ Impossible Date Range

PENALTY:
0.05×

End date before start date — physical impossibility, clearly generated incorrectly

⚠ Career vs Profile YOE

PENALTY: flag

Sum of job durations massively exceeds claimed years_of_experience

⚠ Job Overlap > 1 Year

PENALTY: flag

Two concurrent jobs with >365 day overlap (excluding part-time side gigs)

⚠ Perfect Assessments

PENALTY: 0.6×

≥4 out of 5+ Redrob assessments scored 99-100 — statistically near-impossible

Combined 2+ flags → 0.05× penalty (effective disqualification from top 100)

The Behavioral Availability Multiplier

A perfect-on-paper candidate who hasn't logged in for 6 months isn't actually hireable.

Availability (45% weight)

`open_to_work_flag`

Marked themselves available

`last_active_date`

Recency of platform login

`verified_email / phone`

Reachability verification

`linkedin_connected`

Profile credibility signal

Engagement (35% weight)

`recruiter_response_rate`

% of messages replied to

`avg_response_time_hours`

Speed of response

`interview_completion_rate`

Showed up to interviews

`applications_submitted_30d`

Active job-search signal

Credibility (20% weight)

`profile_completeness_score`

Profile fill-in rate (0-100)

`saved_by_recruiters_30d`

Recruiter bookmark interest

`github_activity_score`

Code contributions (0-100)

`skill_assessment_scores`

Verified platform assessments

Top 10 Candidates

Senior AI Engineer · Pune/Noida, India (Hybrid)

Rank	ID	Title (from profile)	YOE	Location	Score	Key Skills
#1	0018499	Sr ML Engineer	7.2y	Noida, IN	0.999	Weaviate · RecommSys · DeepLearning
#2	0081846	Lead AI Engineer	6.7y	Jaipur, IN	0.965	InfoRetrieval · LlamaIndex · LTR
#3	0079387	AI Engineer	6.9y	Trivandrum, IN	0.934	RecommSys · SentenceTransformers
#4	0046525	Sr ML Engineer	6.1y	Pune, IN	0.921	Elasticsearch · LangChain · ML
#5	0046064	Sr NLP Engineer	8.9y	Coimbatore, IN	0.881	Python · Pinecone · BM25
#6	0071974	Sr AI Engineer	7.8y	India	0.869	LTR · Weaviate · BM25
#7	0064326	Search Engineer	7.6y	India	0.846	Milvus · SemanticSearch · DL
#8	0039754	Sr Applied Sci.	16.2y	India	0.811	FineTuning · LlamaIndex · Qdra
#9	0077337	Staff ML Eng.	7.0y	India	0.811	SemanticSearch · Pinecone · BM
#10	0050454	AI Engineer	6.8y	India	0.798	LangChain · FineTuning · FAISS

Performance & Compute Compliance



53 sec

Total runtime
(100K candidates)



~1,900/s

Throughput
(candidates per second)



<2 GB

Peak RAM
(under 16GB limit)



0

External API calls
(fully offline)

Constraint	Challenge Limit	Our Result	Pass
Runtime limit	≤ 5 minutes	53 seconds	✓
Memory limit	≤ 16 GB RAM	< 2 GB used	✓
Compute type	CPU only, no GPU	Pure Python + numpy	✓
Network access	Offline (no APIs)	Zero external calls	✓
Disk space	≤ 5 GB intermediate	< 10 MB intermediate	✓
Reproducibility	End-to-end in 1 command	python ranker.py candidates.jsonl output.csv	✓

Evidence-Based Reasoning Generation

No hallucination. Every claim traceable to a specific field in the candidate's JSON.

Rank #1

""Senior machine learning engineer with 7.2y experience; prior product-company work at Razorpay; strong on Weaviate, recommendation systems, deep learning; immediate availability (15d notice).""

→ References: current_title, years_of_experience, career_company, specific_skills_by_name, notice_period_days

Rank #17

""Search engineer with 7.8y experience; prior product-company work at CRED; strong on embeddings, Pinecone, Weaviate; long notice period (120d) is a concern.""

→ Honest concern flag: notice period explicitly named. Down-ranked despite strong skills.

Rank #82

""Backend engineer with 6.9y experience; entire career in consulting/services (JD disqualifies pure consulting); inactive for 7mo.""

→ Two compounding negative signals surfaced. Reasoning matches low rank.

Stage 4 checks: specific facts · JD connection · honest concerns · no hallucination · variation · rank consistency

What Sets This Apart



Understands the JD, not just keywords

Explicitly models what the JD means: product company vs consulting, ship vs research bias, behavioral availability. Not a TF-IDF bag-of-words.



Skill depth, not skill count

Each skill scored by proficiency level, peer endorsements, months used, and platform assessment scores. Expert with 0 months = no credit.



Behavioral multiplier on reach-ability

A ghost candidate (great profile, inactive 8 months, 5% response rate) gets 0.3× on their base score. We rank hireable candidates, not resume databases.



Active honeypot detection

Date math, duration consistency checks, and assessment anomaly detection. 2+ flags = 0.05× penalty. We don't just avoid them — we flag them explicitly.



Production-grade performance

53 seconds for 100K candidates. No GPU, no API calls, no precomputation step. Scales to 200K without architectural changes.



Verifiable, non-hallucinating reasoning

Every reasoning sentence cites actual fields from the candidate JSON. Zero invented employers, skills, or experience claims.

How to Reproduce

1

Install dependencies

2

Run the ranker

3

Validate the output

4

Expected result

[illegible]

- Entire ranking logic in one file: `src/ranker.py` · ~400 lines · Pure Python stdlib + numpy

Ranking candidates the way a great recruiter would.

Not by matching keywords. By understanding who actually fits the role — and who can actually be hired.

- ✓ Ranked CSV — 100 candidates, validated
- ✓ ranker.py — single-file, fully reproducible
- ✓ This deck — architecture explained